

Chapter 5

The Variational Approach to Optimal Control Problem

Why Optimal Control?

("300 Years of Optimal Control: from the Brachistochrone to the Maximum Principle," IEEE Control Systems, June 1997, pp. 12-44)

The conventional wisdom holds that optimal control theory was born about 40 years ago in the former Soviet Union, with the work on the "Pontryagin maximum principle" by L. S. Pontryagin and his group. Some mathematicians believe that this new theory was no more than a minor addition to the classical calculus of variations, essentially involving the incorporation of inequality constraints. We believe that optimal control is significantly richer and broader than the calculus of variations, from which it differs in some fundamental ways.

The calculus of variations deals mainly with optimization problems of the following standard form

$$\begin{aligned} \text{Minimize } J &= \int_a^b L(q(t), \dot{q}(t), t) dt \\ \text{subject to } q(a) &= q_a \quad \text{and} \quad q(b) = q_b \end{aligned} \tag{1}$$

Or, equivalently, of the form

$$\begin{aligned} \text{Minimize } J &= \int_a^b L(q(t), u(t), t) dt \\ \text{subject to } q(a) &= q_a \quad \text{and} \quad q(b) = q_b \quad \text{and} \quad \dot{q}(t) = u(t) \\ \text{for } a &\leq t \leq b \end{aligned} \tag{2}$$

The distinctive feature of these problems is that the minimization of (1) takes place in the space of "all" curves, so nothing interesting happens on the level of the set of curves under consideration, and all the nontrivial features of the problem arise because of the Lagrangian L .

Optimal control problems, by contrast, involve a minimization over a set C of curves which is itself determined by some dynamical constraints. For example, C might be the set of all curves $t \rightarrow q(t)$ that satisfy a differential equation

$$\dot{q}(t) = f(q(t), u(t), t) \quad (3)$$

for some choice of the “control function” $t \rightarrow u(t)$. Even more precisely, since it may happen a member of C does not uniquely determine the control u that generates it, we should be talking about trajectory-control pairs $(q(\cdot), u(\cdot))$. So in optimal control problem there are at least two objects that give the situation interesting structure, namely, the dynamics f and the functional J to be minimized. In particular, optimal control theory contains, at the opposite extreme from the calculus of variations, problems where the Lagrangian L is 1, i.e. completely trivial, and therefore all the interesting action occurs because of the dynamics f . Such problems, in which it is desired to minimize time (i.e., the integral J of (2) with $L=1$) among all curves $t \rightarrow q(t)$ that satisfy endpoint constraints as in (2) and are solutions of (3) for some control $t \rightarrow u(t)$, are called *minimum time problems*. It is in these problems that the difference between optimal control and the calculus of variations is most clearly seen, and it is no accident that these were the problems that propelled the development of optimal control in early 1960s, and that time-optimal control is prominently represented in today’s research and in modern optimal control text books.

Bernoulli’s problem (the brachistochrone) is a true minimum time problem of the kind that is studied today in optimal control theory. Moreover, the brachistochrone problem is the first one ever to deal with a dynamical behavior and explicitly ask for the optimal selection of a path. A large part of the subsequent history of the calculus of variations can be best understood as the search for simplest and most general statement of the necessary conditions for optimality, and this statement is provided by *the maximum principle of optimal control theory*.

5.1 Necessary conditions for optimal control

To find an admissible control $\bar{u}^*(t)$ that causes the system

$$\dot{\bar{x}}(t) = \bar{f}(\bar{x}, \bar{u}, t)$$

to follow an admissible trajectory $\bar{x}^*(t)$ that extrimizes the performance measure

$$J(\bar{u}) = h(\bar{x}(t_f), t_f) + \int_{t_0}^{t_f} g(\bar{x}(t), \bar{u}(t), t) dt$$

$\bar{x}(t_0) = \bar{x}_0$ and $\bar{x}(t_f) = \bar{x}_f$ are given, t_f is fixed.

$$\bar{x} \in \mathbf{R}^n, \quad \bar{u} \in \mathbf{R}^m, \quad m \leq n$$

$\bar{u}$ is unconstrained

$\bar{x}$ is also unconstrained inside (t_0, t_f)

The augmented functional

$$J_a = h(\bar{x}(t_f), t_f) + \int_{t_0}^{t_f} [g + \bar{\Lambda}^T (\bar{f} - \dot{\bar{x}})] dt \quad \bar{\Lambda}(t); \text{ Lagrange multipliers}$$

$$\bar{\Lambda} \in \mathbf{R}^n$$

$$\delta J_a = \frac{\partial h}{\partial \bar{x}(t_f)} \delta \bar{x}(t_f) + \int_{t_0}^{t_f} \left[\frac{\partial g}{\partial \bar{x}} \delta \bar{x} + \frac{\partial g}{\partial \bar{u}} \delta \bar{u} + \bar{\Lambda}^T \left\{ \frac{\partial \bar{f}}{\partial \bar{x}} \delta \bar{x} + \frac{\partial \bar{f}}{\partial \bar{u}} \delta \bar{u} - \delta \dot{\bar{x}} \right\} + \delta \bar{\Lambda}^T (\bar{f} - \dot{\bar{x}}) \right] dt$$

$$= -\bar{\Lambda}^T \delta \bar{x} \Big|_{t_0}^{t_f} + \int_{t_0}^{t_f} \left[\left(\frac{\partial g}{\partial \bar{x}} + \bar{\Lambda}^T \frac{\partial \bar{f}}{\partial \bar{x}} + \dot{\bar{\Lambda}}^T \right) \delta \bar{x} + \left(\frac{\partial g}{\partial \bar{u}} + \bar{\Lambda}^T \frac{\partial \bar{f}}{\partial \bar{u}} \right) \delta \bar{u} + (\bar{f} - \dot{\bar{x}})^T \delta \bar{\Lambda} \right] dt$$

$$= 0$$

Apply the boundary conditions

The differential equation is also satisfied at all times. $\rightarrow \dot{\bar{x}} = \bar{f}$

For all admissible variations $\delta \bar{x}$ and $\delta \bar{u}$, the necessary conditions are

$$\dot{\bar{\Lambda}}^T = -\frac{\partial g}{\partial \bar{x}} - \bar{\Lambda}^T \frac{\partial \bar{f}}{\partial \bar{x}}$$



; costate equation

$$\frac{\partial g}{\partial \vec{u}} + \vec{\Lambda}^T \frac{\partial \vec{f}}{\partial \vec{u}} = 0$$

$2n$ differential equations ($\dot{\vec{x}}, \dot{\vec{\Lambda}}$) we have $2n$ boundary condition ($\vec{x}(t_0), \vec{x}(t_f)$)

→ Two Point Boundary Value Problem

Define Hamiltonian

$$H \equiv g + \vec{\Lambda}^T \vec{f}, \quad H = H(\vec{x}(t), \vec{u}(t), \vec{\Lambda}(t), t)$$

$$J_a = h(\vec{x}(t_f), t_f) + \int_{t_0}^{t_f} [H - \vec{\Lambda}^T \dot{\vec{x}}] dt$$

$$\begin{aligned} \delta J_a &= \frac{\partial h}{\partial \vec{x}(t_f)} \delta \vec{x}(t_f) + \int_{t_0}^{t_f} \left[\frac{\partial H}{\partial \vec{u}} \delta \vec{u} + \frac{\partial H}{\partial \vec{x}} \delta \vec{x} + \frac{\partial H}{\partial \vec{\Lambda}} \delta \vec{\Lambda} - \delta \vec{\Lambda}^T \dot{\vec{x}} - \vec{\Lambda}^T \delta \dot{\vec{x}} \right] dt \\ &= -\vec{\Lambda}^T \delta \vec{x} \Big|_{t_0}^{t_f} + \int_{t_0}^{t_f} \left[\frac{\partial H}{\partial \vec{u}} \delta \vec{u} + \left(\frac{\partial H}{\partial \vec{x}} + \dot{\vec{\Lambda}}^T \right) \delta \vec{x} + \delta \vec{\Lambda}^T \left(\frac{\partial H}{\partial \vec{\Lambda}} - \dot{\vec{x}} \right) \right] dt \\ &= 0 \end{aligned}$$

Apply the boundary conditions

The differential equation is also satisfied

The condition for $J_a = 0$ for all admissible $\delta \vec{x}, \delta \vec{u}$ are

$$\frac{\partial H}{\partial \vec{u}} = 0 \quad ; \text{ optimality conditions}$$

$$\dot{\vec{\Lambda}} = -\frac{\partial H}{\partial \vec{x}} \quad ; \text{ costate equation (or adjoint equations)}$$

$$\dot{\vec{x}} = \frac{\partial H}{\partial \vec{\Lambda}} = \vec{f} \quad ; \text{ state equations}$$

Example

$$J = \frac{1}{2} \int_0^1 (\dot{x}(t)^2 + u(t)^2) dt$$

$$\dot{x}(t) = u(t)$$

$$x(0) = 1, \quad x(1) = 0$$

$$H = g + \bar{\Lambda}^T \bar{f} = \frac{1}{2} (\dot{x}^2 + u^2) + \Lambda u$$

$$\frac{\partial H}{\partial u} = 0 \quad \rightarrow u + \Lambda = 0$$

$$\therefore u(t) = -\Lambda(t)$$

$$\dot{\Lambda} = -\frac{\partial H}{\partial x} = -x$$

$$\ddot{x} = \dot{u} = -\dot{\Lambda} = x$$

$$x = c_1 e^t + c_2 e^{-t}$$

$$x(0) = 1 \quad \rightarrow c_1 + c_2 = 1$$

$$x(1) = 0 \quad \rightarrow c_1 e + \frac{c_2}{e} = 0$$

$$\Rightarrow c_1 = \frac{1}{1-e^2}$$

$$c_2 = \frac{-e^2}{1-e^2}$$

$$\therefore x(t)^* = \frac{1}{1-e^2} e^t - \frac{e^2}{1-e^2} e^{-t}$$

$$\dot{\Lambda}^* = -x^* = -\frac{1}{1-e^2} e^t + \frac{e^2}{1-e^2} e^{-t}$$

$$u = -\Lambda, \quad \dot{x} = u \quad \Rightarrow \quad \Lambda = -\dot{x},$$

$$u(t) = \dot{x}(t) = \frac{1}{1-e^2} e^t + \frac{e^2}{1-e^2} e^{-t}$$

$$\dot{x} = -\Lambda \quad \text{and} \quad \dot{\Lambda} = -x$$

Note: $\rightarrow \begin{Bmatrix} \dot{x}(t) \\ \dot{\Lambda}(t) \end{Bmatrix} = \begin{bmatrix} 0 & -1 \\ -1 & 0 \end{bmatrix} \begin{Bmatrix} x(t) \\ \Lambda(t) \end{Bmatrix} = A \begin{Bmatrix} x(t) \\ \Lambda(t) \end{Bmatrix}$

Solution is $\begin{Bmatrix} x(t) \\ \Lambda(t) \end{Bmatrix} = e^{At} \begin{Bmatrix} x(0) \\ \Lambda(0) \end{Bmatrix}$

Linear Two Point Boundary Value Problem

$$\begin{bmatrix} \dot{\vec{x}} \\ \dot{\vec{\Lambda}} \end{bmatrix} = A \begin{bmatrix} \vec{x} \\ \vec{\Lambda} \end{bmatrix}$$

A is $2n \times 2n$ constant matrix

$$\vec{x}(t_0) = \vec{x}_0, \quad \vec{x}(t_f) = \vec{x}_f \quad ; 2n \text{ boundary condition given}$$

$$\begin{bmatrix} \vec{x}(t) \\ \vec{\Lambda}(t) \end{bmatrix} = e^{At} \begin{bmatrix} \vec{x}(0) \\ \vec{\Lambda}(0) \end{bmatrix}$$

$$= \begin{bmatrix} \phi_{11}(t) & \phi_{12}(t) \\ \phi_{21}(t) & \phi_{22}(t) \end{bmatrix} \begin{bmatrix} \vec{x}_0 \\ \vec{\Lambda}(0) \end{bmatrix}$$

$$\vec{x}(t) = \phi_{11}(t) \vec{x}_0 + \phi_{12}(t) \vec{\Lambda}(0)$$

$$\vec{x}(t_f) = \vec{x}_f = \phi_{11}(t_f) \vec{x}_0 + \phi_{12}(t_f) \vec{\Lambda}(0)$$

$$\rightarrow \vec{\Lambda}(0) = \phi_{12}^{-1}(t_f) [\vec{x}_f - \phi_{11}(t_f) \vec{x}_0]$$

$$\Rightarrow \begin{pmatrix} \vec{x}(t) \\ \vec{\Lambda}(t) \end{pmatrix} = e^{At} \begin{pmatrix} \vec{x}_0 \\ \vec{\Lambda}(0) \end{pmatrix}$$

$\Rightarrow$ Control $u(t)$ can be calculated using $\vec{\Lambda}(t)$ and $\vec{x}(t)$

because equation $\frac{\partial H}{\partial \vec{u}} = 0$ contain $\vec{\Lambda}(t)$ and $\vec{x}(t)$.

< Final time specified, $\vec{x}(t_f)$ free >

$$J(\vec{u}) = h(\vec{x}(t_f), t_f) + \int_{t_0}^{t_f} g(\vec{x}(t), \vec{u}(t), t) dt$$

$$\text{subject to } \dot{\vec{x}}(t) = \vec{f}(\vec{x}(t), \vec{u}(t), t)$$

$$\vec{x}(t_0) = \vec{x}_0, \vec{x}(t_f) = \text{free, and } t_f \text{ is specified}$$

$$H \equiv g + \bar{\Lambda}^T \vec{f} = g + \bar{\Lambda}^T \dot{\vec{x}} = H(\vec{x}(t), \vec{u}(t), \bar{\Lambda}(t), t)$$

$$J_a = h(\vec{x}(t_f), t_f) + \int_{t_0}^{t_f} [H - \bar{\Lambda}^T \dot{\vec{x}}] dt$$

$$\begin{aligned} \delta J_a &= \frac{\partial h}{\partial \vec{x}(t_f)} \delta \vec{x}(t_f) + \int_{t_0}^{t_f} \left[\frac{\partial H}{\partial \vec{u}} \delta \vec{u} + \frac{\partial H}{\partial \vec{x}} \delta \vec{x} + \frac{\partial H}{\partial \bar{\Lambda}} \delta \bar{\Lambda} - \delta \bar{\Lambda}^T \dot{\vec{x}} - \bar{\Lambda}^T \delta \dot{\vec{x}} \right] dt \\ &= \left[\frac{\partial h}{\partial \vec{x}(t_f)} - \bar{\Lambda}^T(t_f) \right] \delta \vec{x}(t_f) + \int_{t_0}^{t_f} \left[\frac{\partial H}{\partial \vec{u}} \delta \vec{u} + \left(\frac{\partial H}{\partial \vec{x}} + \dot{\bar{\Lambda}}^T \right) \delta \vec{x} + \delta \bar{\Lambda}^T \left(\frac{\partial H}{\partial \bar{\Lambda}} - \dot{\vec{x}} \right) \right] dt \\ &= 0 \end{aligned}$$

$$\Rightarrow \frac{\partial H}{\partial \vec{u}} = 0, \quad \dot{\bar{\Lambda}}^T = -\frac{\partial H}{\partial \vec{x}}, \quad \dot{\vec{x}} = \frac{\partial H}{\partial \bar{\Lambda}} = \vec{f}$$

Boundary conditions

$$\bar{\Lambda}^T(t_f) = \frac{\partial h}{\partial \vec{x}} \Big|_{t=t_f} \quad \text{at } t = t_f$$

$$\vec{x}(t_0) = \vec{x}_0$$

Linear Two Point Boundary Value Problem

$$\begin{bmatrix} \dot{\vec{x}}(t) \\ \dot{\bar{\Lambda}}(t) \end{bmatrix} = A \begin{bmatrix} \vec{x}(t) \\ \bar{\Lambda}(t) \end{bmatrix}$$

$$\vec{x}(t_0) = \vec{x}_0, \quad \vec{x}(t_f) = \text{free}$$

$$\Rightarrow \vec{\Lambda}(t_f) = \frac{\partial h^T}{\partial \vec{x}} \Big|_{t=t_f} = \vec{\Lambda}_f$$

$$\begin{aligned} \begin{bmatrix} \vec{x}(t) \\ \vec{\Lambda}(t) \end{bmatrix} &= e^{At} \begin{bmatrix} \vec{x}(0) \\ \vec{\Lambda}(0) \end{bmatrix} \\ &= \begin{bmatrix} \phi_{11}(t) & \phi_{12}(t) \\ \phi_{21}(t) & \phi_{22}(t) \end{bmatrix} \begin{bmatrix} \vec{x}_0 \\ \vec{\Lambda}(0) \end{bmatrix} \end{aligned}$$

$$\vec{\Lambda}(t_f) = \vec{\Lambda}_f = \phi_{21}(t_f) \vec{x}_0 + \phi_{22}(t_f) \vec{\Lambda}(0)$$

$$\rightarrow \vec{\Lambda}(0) = \phi_{22}^{-1}(t_f) [\vec{\Lambda}_f - \phi_{21}(t_f) \vec{x}_0]$$

$$\therefore \vec{\Lambda}(t) = e^{At} \vec{\Lambda}(0)$$

Generally,

$$\dot{\vec{x}}(t) = \vec{f}(\vec{x}(t), \vec{u}(t), t), \quad \vec{x} \in \mathbb{R}^n$$

$$J(x) = h(\vec{x}(t_f), t_f) + \int_{t_0}^{t_f} g(\vec{x}(t), \vec{u}(t), t) dt$$

$$\vec{x}(0) = \vec{x}_0 \text{ (given)}$$

$$\vec{x}_i(t_f) = \vec{x}_f \text{ (given)}, \quad i = 1, 2, \dots, k$$

$$\vec{x}_j(t_f) = \text{free}, \quad j = k+1, k+2, \dots, n$$

If $x_j(t_f)$ = free, then

$$\Rightarrow \vec{\Lambda}_j(t_f) = \frac{\partial h^T}{\partial \vec{x}_j} \Big|_{t=t_f}$$

If all other final states are fixed,

$$x_i(t_f) = x_{if}, \quad i \neq j$$

< **General Optimal Control Problem** >

No inequality constraints on the states and controls

$$J(x) = h(\bar{x}(t_f), t_f) + \int_{t_0}^{t_f} g(\bar{x}, \bar{u}, t) dt$$

$$\text{subject to } \dot{\bar{x}}(t) = \bar{f}(\bar{x}, \bar{u}, t)$$

$$\bar{x}(t_0) = \text{given},$$

$$t_f \text{ and } \bar{x}(t_f) \text{ are free}$$

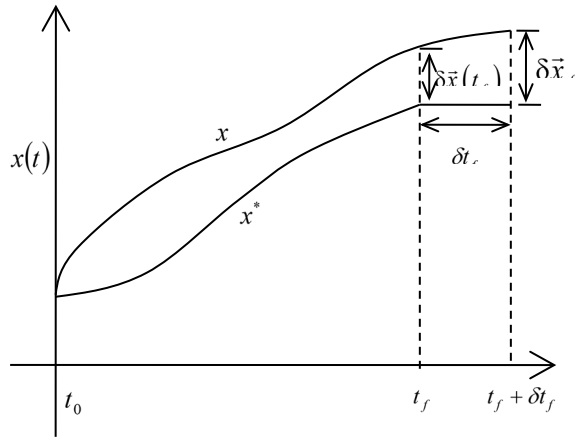
$$\bar{m}(\bar{x}(t_f), t_f) = 0 \quad m \in \mathbf{R}^k; \quad \text{terminal conditions}$$

$$J_a = h(\bar{x}(t_f), t_f) + \bar{v}(t)^T \bar{m}(\bar{x}(t_f), t_f) + \int_{t_0}^{t_f} (H(\bar{x}(t), \bar{u}(t), \bar{\lambda}(t), t) - \bar{\lambda}(t)^T \dot{\bar{x}}(t)) dt$$

$$\text{where } H \equiv g + \bar{\lambda}^T \bar{f}$$

$$\begin{aligned} \delta J_a = & \frac{\partial h}{\partial \bar{x}(t_f)} \delta \bar{x}_f + \frac{\partial h}{\partial t_f} \delta t_f + \bar{v}^T \frac{\partial \bar{m}}{\partial \bar{x}(t_f)} \delta \bar{x}_f + \bar{v}^T \frac{\partial \bar{m}}{\partial t_f} \delta t_f + \delta \bar{v}^T \bar{m}(\bar{x}(t_f), t_f) \\ & + \int_{t_0}^{t_f} \left[\frac{\partial H}{\partial \bar{u}} \delta \bar{u} + \frac{\partial H}{\partial \bar{x}} \delta \bar{x} + \frac{\partial H}{\partial \bar{\lambda}} \delta \bar{\lambda} - \bar{\lambda}^T \delta \dot{\bar{x}} - \delta \bar{\lambda}^T \dot{\bar{x}} \right] dt \\ & + \int_{t_f}^{t_f + \delta t_f} (H - \bar{\lambda}^T \dot{\bar{x}}) dt \end{aligned}$$

$$\int_{t_f}^{t_f + \delta t_f} (H - \bar{\lambda}^T \dot{\bar{x}}) dt \approx (H - \bar{\lambda}^T \dot{\bar{x}}) \Big|_{t_f^*} \delta t_f$$



$$\delta \bar{x}(t_f) = \delta \bar{x}_f - \dot{\bar{x}}^*(t_f^*) \delta t_f$$

$$\begin{aligned}
\delta J_a &= \frac{\partial h}{\partial \bar{x}(t_f)} \delta \bar{x}_f + \frac{\partial h}{\partial t_f} \delta t_f + \bar{v}^T \frac{\partial \bar{m}}{\partial \bar{x}(t_f)} \delta \bar{x}_f + \bar{v}^T \frac{\partial \bar{m}}{\partial t_f} \delta t_f - \bar{\lambda}^T \delta \bar{x}(t_f) \\
&\quad + \int_{t_0}^{t_f} \left[\frac{\partial H}{\partial \bar{u}} \delta \bar{u} + \frac{\partial H}{\partial \bar{x}} \delta \bar{x} + \dot{\bar{\lambda}}^T \delta \bar{x} + \delta \bar{\lambda}^T \left(\frac{\partial H^T}{\partial \bar{\lambda}} - \dot{\bar{x}} \right) \right] dt \\
&\quad + \left[H(t_f^*) - \bar{\lambda}^T(t_f^*) \dot{\bar{x}}(t_f^*) \right] \delta t_f \\
&= \left[\frac{\partial h}{\partial \bar{x}(t_f)} + \bar{v}^T \frac{\partial \bar{m}}{\partial \bar{x}(t_f)} - \bar{\lambda}^T(t_f) \right] \delta \bar{x}_f \\
&\quad + \left[\frac{\partial h}{\partial t_f} + \bar{v}^T \frac{\partial \bar{m}}{\partial t_f} + \bar{\lambda}^T(t_f) \dot{\bar{x}}(t_f) + H(t_f) - \bar{\lambda}^T(t_f) \dot{\bar{x}}(t_f) \right] \delta t_f \\
&\quad + \int_{t_0}^{t_f} \left[\frac{\partial H}{\partial \bar{u}} \delta \bar{u} + \left(\frac{\partial H}{\partial \bar{x}} + \dot{\bar{\lambda}}^T \right) \delta \bar{x} + \delta \bar{\lambda}^T \left(\frac{\partial H^T}{\partial \bar{\lambda}} - \dot{\bar{x}} \right) \right] dt \\
&= 0
\end{aligned}$$

Necessary conditions

$$\frac{\partial H}{\partial \bar{u}} = \bar{0}$$

optimality conditions

$$\dot{\bar{\lambda}} = - \frac{\partial H^T}{\partial \bar{x}}$$

costate (adjoint) equations

$$\dot{\bar{x}} = \frac{\partial H^T}{\partial \bar{\lambda}} = \bar{f}$$

state equations

$$\bar{\lambda}^T(t_f) = \left[\frac{\partial h}{\partial \bar{x}(t_f)} + \bar{v}^T \frac{\partial \bar{m}}{\partial \bar{x}(t_f)} \right]_{t=t_f}$$

boundary conditions

$$H(t_f) = - \left[\frac{\partial h}{\partial t_f} + \bar{v}^T \frac{\partial \bar{m}}{\partial t_f} \right]_{t=t_f}$$

boundary conditions

$$\bar{x}(t_0) = \bar{x}_0 \quad \text{given}$$

boundary conditions

$\bar{x}(t_0)$ are given, $\bar{\lambda}(t_f)$ is determined by boundary conditions. $\bar{\lambda}(t)$ must evidently be determined as an intermediate step in finding the optimal control $\bar{u}^*(t)$ which depend on $\bar{\lambda}(t)$ through the optimality conditions $(\frac{\partial H}{\partial \bar{u}} = \bar{0})$.

< Hamiltonian >

The time derivative of the Hamiltonian;

$$\begin{aligned}\frac{dH}{dt} &= \frac{\partial H}{\partial t} + \frac{\partial H}{\partial \bar{x}} \frac{d\bar{x}}{dt} + \frac{\partial H}{\partial \bar{u}} \frac{d\bar{u}}{dt} + \frac{\partial H}{\partial \bar{\Lambda}} \frac{d\bar{\Lambda}}{dt} \\ &= H_t + H_{\bar{u}} \dot{\bar{u}} + \left(H_{\bar{x}} + \dot{\bar{\Lambda}}^T \right) \bar{f}\end{aligned}$$

$$H(\bar{x}, \bar{u}, \bar{\Lambda}, t)$$

$$H = g + \bar{\Lambda}^T \cdot \bar{f}$$

$$\bar{f} = \dot{\bar{x}}$$

If $\bar{u}(t)$ is an optimal control $H_{\bar{u}}=0$, $\dot{\bar{\Lambda}}^T = -H_{\bar{x}}$

$$\dot{H} = H_t$$

Now, in the time-invariant case, f and g are not explicit functions of t , and so neither is H .

In this situation,

$$\dot{H} = 0$$

Hence, for time-invariant systems and cost functions, the Hamiltonian is a constant on the optimal solution.

If t_f is free (or minimum time problem) and $h=0$, $\mathbf{m}=0$

$$H(t_f) = - \left[\frac{\partial h}{\partial t_f} + \bar{v}^T \frac{\partial \bar{m}}{\partial t_f} \right]_{t=t_f} = 0$$

$$\dot{H} = 0 \Rightarrow H(t) = c \text{ and } H(t_f) = 0 \Rightarrow H(t) = 0$$

SUMMARY OF BOUNDARY CONDITIONS IN OPTIMAL CONTROL
(Table 5-1)

Problem	Description	Substitution	Boundary -condition equations	Remark
t_f fixed	1. $\mathbf{x}(t_f) = \mathbf{x}_f$ specified final state	$\delta \mathbf{x}_f = \delta \mathbf{x}(t_f) = \mathbf{0}$ $\delta t_f = 0$	$\mathbf{x}^*(t_0) = \mathbf{x}_0$ $\mathbf{x}^*(t_f) = \mathbf{x}_f$	$2n$ equations to determine $2n$ constraints of integration
	2. $\mathbf{x}(t_f)$ free	$\delta \mathbf{x}_f = \delta \mathbf{x}(t_f)$ $\delta t_f = 0$	$\mathbf{x}^*(t_0) = \mathbf{x}_0$ $\frac{\partial h}{\partial \mathbf{x}}(\mathbf{x}^*(t_f)) - \boldsymbol{\Lambda}^*(t_f) = \mathbf{0}$	$2n$ equations to determine $2n$ constraints of integration
	3. $\mathbf{x}(t_f)$ on the surface $\mathbf{m}(\mathbf{x}(t)) = 0$	$\delta \mathbf{x}_f = \delta \mathbf{x}(t_f)$ $\delta t_f = 0$	$\mathbf{x}^*(t_0) = \mathbf{x}_0$ $\frac{\partial h}{\partial \mathbf{x}}(\mathbf{x}^*(t_f)) - \boldsymbol{\Lambda}^*(t_f) = \sum_{i=1}^k \nu_i \left[\frac{\partial m_i}{\partial \mathbf{x}}(\mathbf{x}^*(t_f)) \right]$ $\mathbf{m}(\mathbf{x}^*(t_f)) = \mathbf{0}$	$(2n+k)$ equations to determine $2n$ constraints of integration and the variables $\nu_1, \dots, \nu_k$
t_f free	4. $\mathbf{x}(t_f) = \mathbf{x}_f$ specified final state	$\delta \mathbf{x}_f = \mathbf{0}$	$\mathbf{x}^*(t_0) = \mathbf{x}_0$ $\mathbf{x}^*(t_f) = \mathbf{x}_f$ $H(\mathbf{x}^*(t_f), \mathbf{u}(t_f), \boldsymbol{\Lambda}^*(t_f), t_f) + \frac{\partial h}{\partial t}(\mathbf{x}^*(t_f), t_f) = 0$	$(2n+1)$ equations to determine $2n$ constraints of integration t_f
	5. $\mathbf{x}(t_f)$ free		$\mathbf{x}^*(t_0) = \mathbf{x}_0$ $\frac{\partial h}{\partial \mathbf{x}}(\mathbf{x}^*(t_f)) - \boldsymbol{\Lambda}^*(t_f) = \mathbf{0}$ $H(\mathbf{x}^*(t_f), \mathbf{u}^*(t_f), \boldsymbol{\Lambda}^*(t_f), t_f) + \frac{\partial h}{\partial t}(\mathbf{x}^*(t_f), t_f) = 0$	$(2n+1)$ equations to determine $2n$ constraints of integration t_f
	6. $\mathbf{x}(t_f)$ on the moving $\boldsymbol{\theta}(t)$	$\delta \mathbf{x}_f = \frac{d\boldsymbol{\theta}}{dt}(t_f) \delta t_f$	$\mathbf{x}^*(t_0) = \mathbf{x}_0$ $\mathbf{x}^*(t_f) = \boldsymbol{\theta}(t_f)$ $H(\mathbf{x}^*(t_f), \mathbf{u}^*(t_f), \boldsymbol{\Lambda}^*(t_f), t_f) + \frac{\partial h}{\partial t}(\mathbf{x}^*(t_f), t_f) + \left[\frac{\partial h}{\partial \mathbf{x}}(\mathbf{x}^*(t_f), t_f) - \boldsymbol{\Lambda}^*(t_f) \right]^T \frac{d\boldsymbol{\theta}}{dt}(t_f) = 0$	$(2n+1)$ equations to determine $2n$ constraints of integration t_f
	7. $\mathbf{x}(t_f)$ on the surface $\mathbf{m}(\mathbf{x}(t)) = \mathbf{0}$		$\mathbf{x}^*(t_0) = \mathbf{x}_0$ $\frac{\partial h}{\partial \mathbf{x}}(\mathbf{x}^*(t_f)) - \boldsymbol{\Lambda}^*(t_f) = \sum_{i=1}^k \nu_i \left[\frac{\partial m_i}{\partial \mathbf{x}}(\mathbf{x}^*(t_f)) \right]$ $\mathbf{m}(\mathbf{x}(t_f)) = \mathbf{0}$ $H(\mathbf{x}^*(t_f), \mathbf{u}^*(t_f), \boldsymbol{\Lambda}^*(t_f), t_f) + \frac{\partial h}{\partial t}(\mathbf{x}^*(t_f), t_f) = 0$	$(2n+k+1)$ equations to determine $2n$ constraints of integration, the variables $\nu_1, \dots, \nu_k$ and t_f
	8. $\mathbf{x}(t_f)$ on the		$\mathbf{x}^*(t_0) = \mathbf{x}_0$	$(2n+k+1)$ equations to

	surface $\mathbf{m}(\mathbf{x}(t), t) = \mathbf{0}$	$\frac{\partial h}{\partial \mathbf{x}}(\mathbf{x}^*(t_f)) - \boldsymbol{\Lambda}^*(t_f) = \sum_{i=1}^k \nu_i \left[\frac{\partial m_i}{\partial \mathbf{x}}(\mathbf{x}^*(t_f), t_f) \right]$ $\mathbf{m}(\mathbf{x}(t_f), t_f) = \mathbf{0}$ $H(\mathbf{x}^*(t_f), \mathbf{u}^*(t_f), \boldsymbol{\Lambda}^*(t_f), t_f) + \frac{\partial h}{\partial t}(\mathbf{x}^*(t_f), t_f)$ $= \sum_{i=1}^k \nu_i \left[\frac{\partial m_i}{\partial t}(\mathbf{x}^*(t_f), t_f) \right]$	determine $2n$ constraints of integration, the variables $\nu_1, \dots, \nu_k$, and t_f
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Significance of the initial costate values

$\bar{\Lambda}(0) = ?$ What do these numbers tell us?

$$J(\mathbf{x}) = \int_{t_0}^{t_f} g(\bar{\mathbf{x}}, \bar{\mathbf{u}}, t) dt$$

$$\dot{\bar{\mathbf{x}}}(t) = \bar{\mathbf{f}}(\bar{\mathbf{x}}, \bar{\mathbf{u}}, t)$$

$\bar{\mathbf{x}}(t_0)$; unspecified, t_0 is fixed

$\bar{\mathbf{x}}(t_f)$; specified, t_f is fixed

$$H \equiv g + \bar{\Lambda}^T \bar{\mathbf{f}}$$

$$J_a = \int_{t_0}^{t_f} (H - \bar{\Lambda}^T \dot{\bar{\mathbf{x}}}) dt$$

$$\delta J_a = \int_{t_0}^{t_f} \left[\frac{\partial H}{\partial \bar{\mathbf{u}}} \delta \bar{\mathbf{u}} + \frac{\partial H}{\partial \bar{\mathbf{x}}} \delta \bar{\mathbf{x}} + \frac{\partial H}{\partial \bar{\Lambda}} \delta \bar{\Lambda} - \dot{\bar{\Lambda}}^T \delta \dot{\bar{\mathbf{x}}} - \delta \bar{\Lambda}^T \dot{\delta \bar{\mathbf{x}}} \right] dt$$

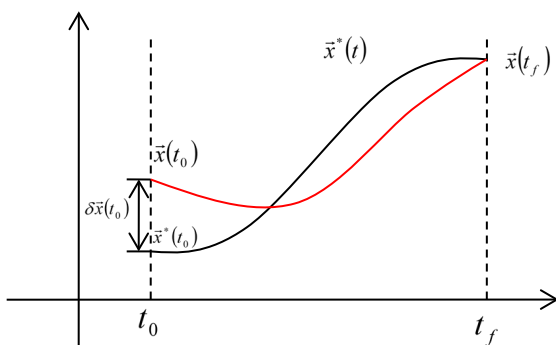
$$= -\bar{\Lambda}^T \delta \bar{\mathbf{x}}|_{t_0}^{t_f} + \int_{t_0}^{t_f} \left[\frac{\partial H}{\partial \bar{\mathbf{u}}} \delta \bar{\mathbf{u}} + \left(\frac{\partial H}{\partial \bar{\mathbf{x}}} + \dot{\bar{\Lambda}}^T \right) \delta \bar{\mathbf{x}} + \delta \bar{\Lambda}^T \left(\frac{\partial H}{\partial \bar{\Lambda}} - \dot{\bar{\mathbf{x}}} \right) \right] dt$$

$$= \bar{\Lambda}(t_0)^T \delta \bar{\mathbf{x}}(t_0) + \int_{t_0}^{t_f} \left[\frac{\partial H}{\partial \bar{\mathbf{u}}} \delta \bar{\mathbf{u}} + \left(\frac{\partial H}{\partial \bar{\mathbf{x}}} + \dot{\bar{\Lambda}}^T \right) \delta \bar{\mathbf{x}} + \delta \bar{\Lambda}^T \left(\frac{\partial H}{\partial \bar{\Lambda}} - \dot{\bar{\mathbf{x}}} \right) \right] dt$$

$$= 0$$

$$\Rightarrow \begin{aligned} H_u &= 0 \\ \dot{\bar{\lambda}} &= -H_{\bar{x}} \\ \dot{\bar{x}} &= H_{\bar{\lambda}} \end{aligned}$$

$$\delta J_a = \bar{\lambda}(t_0)^T \delta \bar{x}(t_0) \quad \text{transversality condition}$$



$$\bar{x}^*(t_0) \rightarrow J^*$$

$$\bar{x}^*(t_0) + \delta \bar{x}(t_0) \rightarrow J^* + \delta J$$

$$\delta J = \delta J_a \quad \text{since all equations are satisfied.}$$

$$\delta J_a = \bar{\lambda}(t_0)^T \delta \bar{x}(t_0) \rightarrow \bar{\lambda}(0) = \frac{\partial J}{\partial \bar{x}(0)}$$

$\bar{\lambda}(0)$ tells us that the sensitivity of the cost with respect to initial conditions.

5.2 Linear Quadratic Problem (LQR)

Many dynamic systems of interest are well approximated as linear. If quadratic performance indices and linear state equations are used to pose optimal problems for linear system, the optimal controls are linear feedback on the state variables.

$$\text{Minimize } J = \frac{1}{2} \vec{x}^T(t_f) L \vec{x}(t_f) + \frac{1}{2} \int_{t_0}^{t_f} [\vec{x}^T(t) Q(t) \vec{x}(t) + \vec{u}^T(t) R(t) \vec{u}(t)] dt$$

$$\text{subject to } \dot{\vec{x}}(t) = A(t) \vec{x}(t) + B(t) \vec{u}(t)$$

$$\vec{x} \in \Re^n, \quad \vec{u} \in \Re^m \quad A: (n \times n), B: (n \times m)$$

$$\vec{x}(t_0) = \vec{x}_0, \quad \vec{x}(t_f) = \text{free}, \quad t_f \text{ is fixed.}$$

Q is symmetric positive semi-definite⁴ matrix ($n \times n$)

R is symmetric positive definite matrix ($m \times m$)

L is symmetric positive semi-definite matrix ($n \times n$)

The control and states are not bounded.

It is desired to maintain the state vector close to the origin without an excessive expenditure of control effort.

The Hamiltonian is

⁴ Symmetric A matrix such that $A \in \mathbf{R}^{n \times n}$ is positive definite if $\vec{x}^T A \vec{x} > 0$ for all non-zero $\vec{x} \in \mathbf{R}^n$, and symmetric A matrix such that $A \in \mathbf{R}^{n \times n}$ is positive semi-definite if $\vec{x}^T A \vec{x} \geq 0$ for all non-zero $\vec{x} \in \mathbf{R}^n$.

$$H = \frac{1}{2} \bar{x}^T Q \bar{x} + \frac{1}{2} \bar{u}^T R \bar{u} + \bar{\Lambda}^T (A \bar{x} + B \bar{u})$$

$$\dot{\bar{\Lambda}} = - \left(\frac{\partial H}{\partial \bar{x}} \right)^T = -A^T \bar{\Lambda} - Q \bar{x}$$

$$\frac{\partial H}{\partial \bar{u}} = 0 \Rightarrow R \bar{u} + B^T \bar{\Lambda} = 0$$

$$\Rightarrow \bar{u} = -R^{-1} B^T \bar{\Lambda}, \quad \text{The existence of } R^{-1} \text{ is assured because } R \text{ is a positive definite matrix}$$

$$\dot{\bar{x}} = A \bar{x} + B \bar{u} = A \bar{x} - B R^{-1} B^T \bar{\Lambda}, \quad \bar{x}(t_0) = \bar{x}_0, \quad \bar{\Lambda}(t_f) = \frac{\partial h}{\partial \bar{x}} \Big|_{t=t_f} = L \bar{x}(t_f)$$

Kalman define $\bar{\Lambda}(t) = [K(t)] \bar{x}(t)$, $\bar{\Lambda}(t)$ is linear function of the states.

$$\begin{aligned} \dot{\bar{\Lambda}} &= \dot{K} \bar{x} + K \dot{\bar{x}} & \dot{\bar{\Lambda}} &= -A^T \bar{\Lambda} - Q \bar{x} = -A^T K \bar{x} - Q \bar{x} \\ &= \dot{K} \bar{x} + K (A \bar{x} - B R^{-1} B^T \bar{\Lambda}) \end{aligned}$$

$$\rightarrow \dot{K} \bar{x} + K (A \bar{x} - B R^{-1} B^T K \bar{x}) = -A^T K \bar{x} - Q \bar{x}$$

$$\rightarrow [\dot{K} + K A - K B R^{-1} B^T K + A^T K + Q] \bar{x} = 0$$

This equation must be satisfied at all times.

$$\dot{K}(t) = -K(t)A(t) - A^T(t)K(t) - Q(t) + K(t)B(t)R^{-1}(t)B^T(t)K(t) \quad ; \text{Riccati differential equation}$$

Boundary conditions

$$\bar{\Lambda}(t_f) = L \bar{x}(t_f) \quad \rightarrow \quad K(t_f) \bar{x}(t_f) = L \bar{x}(t_f)$$

$$\rightarrow K(t_f) = L \quad ; \text{requires backward integration}$$

K is n x n matrix $\rightarrow$ the system is n^2 first-order differential equations.

K is symmetric $\rightarrow$ not n^2 equation, but $\frac{n(n+1)}{2}$ first-order differential equations.

These equation can be integrated numerically.

The integration is started at $t=t_f$ and process backward in time to at $t=t_0$.

$K(t)$ is stored and $\bar{u}(t)$ is calculated as

$$\begin{aligned}\bar{u}(t) &= -R^{-1}B^T\bar{\Lambda}(t) \\ &= -R^{-1}B^TK(t)\bar{x}(t), \quad \leftarrow \bar{x}(t) \text{ measured or computed}\end{aligned}$$

Note $\dot{\bar{x}} = A\bar{x} + B\bar{u} = A\bar{x}(t) - BR^{-1}B^TK(t)\bar{x}(t)$, $\bar{x}(t_0) = \bar{x}_0$ (given)

integration forward $t=t_0$ to $t=t_f$. $\in \mathfrak{R}^n \Rightarrow \bar{x}(t)$

This indicated that the optimal control law is a linear, time-varying, combination of the system states.

< *Special case of LQR* >

If Q, R, A, B are constant(time-invariant) and $L=0$, t_f is free, $\bar{x}(t_f)$ is free,

Transversality conditions

$$\left(\frac{\partial h}{\partial \bar{x}} + \mathbf{v}^T \frac{\partial \bar{m}}{\partial \bar{x}} - \bar{\Lambda} \right)_{t=t_f} \delta \bar{x}_f + \left(\frac{\partial h}{\partial t} + \mathbf{v}^T \frac{\partial \bar{m}}{\partial t} + H \right)_{t=t_f} \delta t_f = 0$$

$$\bar{\Lambda}(t_f) = 0, \quad H(t_f) = 0$$

What can we say if the Hamiltonian does not explicitly depend on time.

$$H(t) = \text{constant}, \quad \text{and} \quad H(t_f) = 0$$

$$\Rightarrow H(t) = 0 = \frac{1}{2} \bar{x}^T Q \bar{x} + \frac{1}{2} \bar{u}^T R \bar{u} + \bar{\Lambda}^T (A\bar{x} + B\bar{u})$$

$$\bar{u} = -R^{-1}B^T\bar{\Lambda} = -R^{-1}B^TK\bar{x}$$

$$\bar{\Lambda} = K\bar{x} \rightarrow \bar{\Lambda}^T = \bar{x}^TK^T = \bar{x}^TK$$

K and R are symmetric matrix

$$= \frac{1}{2} \bar{x}^T \left[Q + K^T B R^{-1} R B^T K + 2KA - 2KBR^{-1}B^TK \right] \bar{x}$$

$$\therefore Q + 2KA - KBR^{-1}B^TK = 0$$

algebraic Riccati equation

$$\Rightarrow K(t)$$

$$\Rightarrow \bar{u}(t) = -R^{-1}B^T K(t)\bar{x}(t) \quad \text{with} \quad \bar{x}(t)$$

Note $\dot{K}(t) = 0 \Leftrightarrow K(t) = \text{const.}$ as $t_f \rightarrow \infty$

< Symetricity of R, Q, L and K matrix >

Example)

$$x_1^2 + x_1 x_2 + x_2^2, \quad \bar{x} = \begin{Bmatrix} x_1 \\ x_2 \end{Bmatrix}$$

$$\rightarrow \text{Let } Q = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}, \quad \bar{x}^T Q \bar{x} = \begin{pmatrix} x_1 & x_2 \end{pmatrix} \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \end{Bmatrix} = x_1^2 + x_1 x_2 + x_2^2$$

$$\text{let } Q = \begin{bmatrix} 1 & \frac{1}{2} \\ \frac{1}{2} & 1 \end{bmatrix}, \quad \bar{x}^T Q \bar{x} = \begin{pmatrix} x_1 & x_2 \end{pmatrix} \begin{bmatrix} 1 & \frac{1}{2} \\ \frac{1}{2} & 1 \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \end{Bmatrix} = x_1^2 + x_1 x_2 + x_2^2$$

Thus, Q matrix can be chosen as a symmetric matrix.

Similarly, R or L matrix can be chosen as a symmetric matrix.

The Riccati Equation:

$$\dot{K}(t) = -K(t)A(t) - A^T(t)K(t) - Q(t) + K(t)B(t)R^{-1}(t)B^T(t)K(t) \quad (*)$$

$$K(t_f) = L$$

Taking the transpose of the Riccati equation gives

$$[\dot{K}(t)]^T = [-K(t)A(t) - A^T(t)K(t) - Q(t) + K(t)B(t)R^{-1}(t)B^T(t)K(t)]^T$$

Let's omit the argument (t) for simplicity.

By using the matrix identity $[CD]^T = D^T C^T$, we have

$$[\dot{K}]^T = -A^T K^T - K^T A - Q^T + K^T B [R^{-1}]^T B^T K^T$$

Note that Q is symmetric; R is symmetric, so R^{-1} is symmetric and $[R^{-1}]^T = R^{-1}$.
Therefore,

$$[\dot{K}]^T = -A^T K^T - K^T A - Q + K^T B R^{-1} B^T K^T \quad (1)$$

We know that operations of transposition and differentiation are commutative;

$$\text{i.e. } \left[\frac{d}{dt} K \right]^T = \frac{d}{dt} [K^T]$$

Then we have

$$\frac{d}{dt} [K^T] = -A^T K^T - K^T A - Q + K^T B R^{-1} B^T K^T \quad (2)$$

The boundary condition for eq(1) is

$$K^T(t_f) = L^T = L$$

In eq(2), let $K^T \equiv M$, then

$$\begin{aligned} \dot{M} &= -A^T M - M A - Q + M B R^{-1} B^T M \\ M(t_f) &= L \end{aligned} \quad (3)$$

Note that M(t) in eq(3) and K(t) in the Riccati equation(eq. *) satisfy the same differential equation with the same boundary conditions.

Therefore, since the solution is unique, we conclude that

$$M(t) \equiv K^T(t) = K(t)$$

Since K(t) matrix is symmetric, we need solve only for the elements of K(t) which lie on or above the main diagonal. The number of terms involved is $1+2+3+\dots+n$. This sum is equal to $n(n+1)/2$.

< Linear Tracking Problem >

Let us generalize the results obtained for the linear regulator problem to the tracking problem:

If initial condition is changed a little bit.

How to control the system about reference solution.

The state equations are

$$\dot{\bar{x}}(t) = A(t)\bar{x}(t) + B(t)\bar{u}(t)$$

$\bar{x}(t_0)$ is given

$\bar{x}(t_f)$ is free, t_f is fixed.

The performance measure to be minimized is

$$J = \frac{1}{2} [\bar{x}(t_f) - \bar{r}(t_f)]^T L [\bar{x}(t_f) - \bar{r}(t_f)] \\ + \frac{1}{2} \int_{t_0}^{t_f} [\bar{x}(t) - \bar{r}(t)]^T Q(t) [\bar{x}(t) - \bar{r}(t)] + \bar{u}^T(t) R(t) \bar{u}(t) dt$$

$\bar{r}(t_f)$ is known or calculated prior

$\bar{r}(t)$ is the desired or reference value of the state vector.

The Hamiltonian is

$$H = \frac{1}{2} \{ (\bar{x} - \bar{r})^T Q (\bar{x} - \bar{r}) + \bar{u}^T R \bar{u} \} + \bar{\lambda}^T (A\bar{x} + B\bar{u})$$

$$\frac{\partial H}{\partial \bar{u}} = \bar{0} \rightarrow R\bar{u} + \bar{\lambda}^T B = \bar{0}, \quad \bar{u} = -R^{-1} B^T \bar{\lambda}$$

$$\dot{\bar{x}} = A\bar{x} + B\bar{u} = A\bar{x} - B R^{-1} B^T \bar{\lambda}$$

$$\dot{\bar{\lambda}} = - \left(\frac{\partial H}{\partial \bar{x}} \right)^T = -Q(\bar{x} - \bar{r}) - A^T \bar{\lambda} = -Q\bar{x} - A^T \bar{\lambda} + Q\bar{r}$$

Transversality condtion

$$\left(\frac{\partial h}{\partial \vec{x}} + \vec{v}^T \frac{\partial \vec{m}}{\partial \vec{x}} - \vec{\Lambda} \right)_{t=t_f} \delta x_f + \left(\frac{\partial h}{\partial t} + \vec{v}^T \frac{\partial \vec{m}}{\partial t} + H \right)_{t=t_f} \delta t_f = 0$$

$$\vec{\Lambda}(t_f) = \frac{\partial h}{\partial \vec{x}} \Big|_{t_f} = L[\vec{x}(t_f) - \vec{r}(t_f)]$$

Let $\vec{\Lambda}(t) = K(t)\vec{x}(t) + S(t)$

$$\begin{aligned} \dot{\vec{\Lambda}} &= \dot{K}\vec{x} + K\dot{\vec{x}} + \dot{S} \\ &= \dot{K}\vec{x} + K[A\vec{x} - BR^{-1}B^T(K\vec{x} + S)] + \dot{S} \end{aligned}$$

$$\rightarrow \dot{K}\vec{x} + K[A\vec{x} - BR^{-1}B^T(K\vec{x} + S)] + \dot{S} = -Q\vec{x} - A^T[K\vec{x} + S] + Q\vec{r}$$

$$\rightarrow [\dot{K} + KA - KBR^{-1}B^TK + A^TK + Q]\vec{x} + [-KBR^{-1}B^TS + \dot{S} + A^TS - Q\vec{r}] = 0$$

The above equation must be satisfied for all $\vec{x}$ and $\vec{r}$.

Necessary conditions

$$\begin{aligned} \dot{K} &= -KA + KBR^{-1}B^TK - A^TK - Q \\ \dot{S} &= -(A^T - KBR^{-1}B^T)S + Q\vec{r} \end{aligned}$$

Boundary conditions

$$\begin{aligned} \vec{\Lambda}(t_f) &= L\vec{x}(t_f) - L\vec{r}(t_f) \\ \vec{\Lambda}(t_f) &= K(t_f)\vec{x}(t_f) + S(t_f) \\ \rightarrow K(t_f) &= L, \quad S(t_f) = -L\vec{r}(t_f) \end{aligned}$$

Note that we know $\vec{r}(t)$ completely prior,

Solve 1) about K

2) about S, using K

$$\therefore \vec{u}(t) = -R^{-1}B^T\vec{\Lambda} = -R^{-1}B^T[K\vec{x} + S]$$

$$\text{if } \vec{r}(t) = 0 \rightarrow S(t) = 0 \quad ; \text{ LQR problem}$$

< State-Dependent Riccati Equation (SDRE) Controller >

The State-Dependent Riccati Equation (SDRE) control method was suggested by James R. Cloutier. The general process of the SDRE technique is the same as that of LQR control method. However, the SDRE technique updates the dynamic model matrices **A** and **B** at each time step because these matrices in the SDRE technique are dependent on the states as shown in Figure 1. It affects the value of gain matrix **K**, so that the controller can make proper control values for the dynamic systems at any time. In order to use the SDRE control method, like LQR control method, the state-space equations like Eq. (14) are needed that can be written as below:

$$\begin{aligned}\dot{\mathbf{x}}(t) &= \mathbf{f}(\mathbf{x}(t)) + \mathbf{g}(\mathbf{x}(t))\mathbf{u}(t) \\ \dot{\mathbf{x}}(t) &= \mathbf{A}(\mathbf{x}(t))\mathbf{x}(t) + \mathbf{B}(\mathbf{x}(t))\mathbf{u}(t)\end{aligned}$$

Where $\mathbf{x}(t)$, $\mathbf{u}(t)$, $\mathbf{A}(\mathbf{x}(t))$ and $\mathbf{B}(\mathbf{x}(t))$ denote states, control, and model matrices, respectively. Note that unlike LQR control method, in the SDRE control method, dynamic model matrices $\mathbf{A}(\mathbf{x}(t))$ and $\mathbf{B}(\mathbf{x}(t))$ are not constant any more. These matrices are dependent on the states.

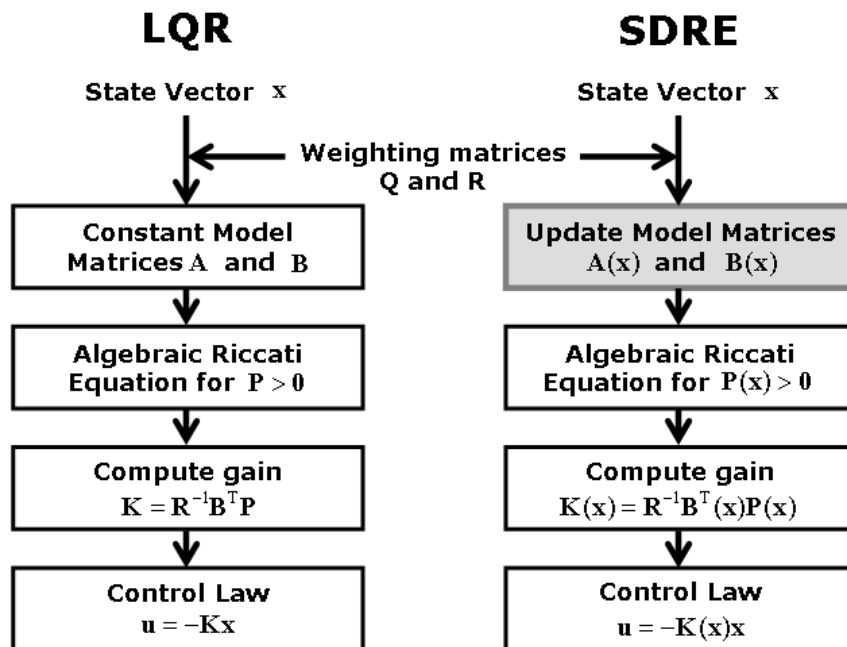


Figure 1 Description of the SDRE technique

Controllability and Observability: A system is said to be controllable at time t_0 if it is possible by means of an unconstrained control vector to transfer the system from any initial state ($\mathbf{x}(t_0)$) to any other state in a finite interval of time. A system is said to be observable at time t_0 if, with the system in state $\mathbf{x}(t_0)$, it is possible to determine this state from the observation of the output over a finite time interval.

Observability is very important because, in practice, the difficulty encountered with state feedback control is that some of the state variables are not accessible for direct measurement, with the result that it becomes necessary to estimate the unmeasurable state variables in order to construct the control signals.

$$\dot{\tilde{\mathbf{x}}}(t) = A\tilde{\mathbf{x}}(t) + B\tilde{\mathbf{u}}(t), \quad \tilde{\mathbf{y}}(t) = C\tilde{\mathbf{x}}(t)$$

The system is completely observable if and only if the (n by nm matrix)

$$\begin{bmatrix} C^T & A^T C^T & \dots & (A^T)^{n-1} C^T \end{bmatrix}$$

is of rank n or has linearly independent column vectors. This matrix is called the observability matrix.

If the system completely state controllable, then, given any initial state $\mathbf{x}(0)$, the rank of the (n by n) matrix

$$\begin{bmatrix} B & AB & \dots & (A)^{n-1} B \end{bmatrix}$$

is of rank n.

Stability is a notion that describes whether the system will be able to follow the input command. In a nonrigorous manner, a system is said to be unstable if its output is out of control or increases without bound.

J. R. Cloutier, "State-Dependent Riccati Equation Techniques: An Overview", *Proceedings of the American Control Conference*, Albuquerque, NM, 1997, pp. 932-936